

Diagnostic

Network meta-analysis for diagnostic tests

10.1136/bmjebm-2019-111179

Jack W O'Sullivan^{1,2,3}

¹Division of Cardiovascular Medicine and Cardiovascular Institute, Department of Medicine, Stanford University, Stanford, California, USA, Stanford University, Palo Alto, California, USA, ²Centre for Evidence-Based Medicine, University of Oxford, Oxford, UK, ³Meta-Research Innovation Center at Stanford (METRICS), Stanford University, Stanford, California, USA

Correspondence to: Dr Jack W O'Sullivan, Division of Cardiology, Department of Medicine, Stanford University, Stanford CA 94305, USA; jackos@stanford.edu

Network meta-analysis (NMA) has enabled the comparison of numerous interventions in the absence of direct head-to-head randomised controlled trials. A new technique describes NMA for diagnostic tests, an approach that can facilitate a similar comparison between different diagnostic tests at different diagnostic thresholds.

Globally, the provision of healthcare is aided by clinical practice guidelines.¹ Healthcare organisations, such as the UK's National Institute for Health and Care Excellence (NICE), routinely provide healthcare workers with advice about what works for which patients in which particular setting.

To do this, NICE, as does the WHO and various American professional societies, conducts systematic reviews and meta-analyses on the available evidence. The recent methodological creation of NMA has enabled these organisations to explicitly compare the effectiveness of different interventions even if they have not been directly compared in the same randomised controlled trial. NMA has facilitated organisations to determine, for instance, that apixaban 5 mg twice daily is the most efficacious novel oral anticoagulant 'for most outcomes' in the prevention of stroke in atrial fibrillation.²

Until recently, the opportunity to compare and determine which diagnostic test provides the optimal diagnostic accuracy for patients in a particular setting has not been available.

Owen *et al*³ have addressed this gap with the development of NMA for diagnostic tests. This statistical advancement enables guideline panels and other users to determine which diagnostic test at which diagnostic threshold has the optimal (a) sensitivity, (b) specificity and (c) the optimal balance between sensitivity and specificity. These terms are defined in the box ("Definition of diagnostic threshold") below.

Owen *et al* illustrate their proposed NMA for diagnostic tests with a worked example. They describe a clinical question a guideline committee might face. They determine which test at which diagnostic threshold has the greatest sensitivity and specificity to diagnose cognitive impairment in patients after a stroke. Their NMA enables them to conclude that the test with the highest sensitivity is the Montreal Cognitive Assessment (MoCA) test at a diagnostic threshold of <26 (where

EBM Verdict

EBM Verdict on: Owen *et al*. Network meta-analysis of diagnostic test accuracy studies identifies and ranks the optimal diagnostic tests and thresholds for health care policy and decision-making. *J Clin Epidemiol* 2018 Jul;99:64-74. doi: 10.1016/j.jclinepi.2018.03.005

► Researchers and guideline committees that wish to compare numerous diagnostic tests and determine which test at which diagnostic thresholds provides the optimal sensitivity and specificity should consider constructing a network meta-analysis for diagnostic tests as outlined by Owen *et al*.

30/30 is perfect cognition); the test with the highest specificity is the Mini-Mental State Exam <25 (where 30/30 is perfect cognition) and MoCA with a threshold of <22 has the optimal balance between sensitivity and specificity.

Prior approaches to meta-analysis of diagnostic accuracy studies have allowed the inclusion of primary diagnostic accuracy studies that used different diagnostic thresholds. These methods account for heterogeneity introduced by different diagnostic thresholds used; however, these conventional methods do not facilitate the explicit identification of an optimal diagnostic threshold.

Guideline committees and researchers that are interested in comparing numerous diagnostic tests to determine which test at what diagnostic threshold is optimal for their clinical question should consider the use of an NMA of diagnostic test accuracy studies. This approach will enable users to determine the optimal test and diagnostic threshold in the diagnosis of a disease of interest for a particular patient group.

Definition of diagnostic threshold

A test's diagnostic threshold refers to a cut-off point in the test's results, above which (typically) the test can be interpreted as a 'positive' result and below which the test is 'negative'. Cut-off points are typically numeric such as a cut-off point of 5 mg/L used in most UK laboratories to indicate a C-reactive protein (CRP) is positive.⁴ A diagnostic test will have a different sensitivity and specificity depending on which diagnostic threshold is chosen. A test with a lower diagnostic threshold (such as a CRP cut-off of 3 mg/L) will have a higher sensitivity (also known as true positive rate), but a lower specificity (also known as true negative rate). Conversely, a test with a higher diagnostic threshold will have a lower sensitivity, but a higher specificity. Another term for diagnostic threshold is the reference range, that is, the reference range for CRP is 0–5 mg/L. A diagnostic threshold with the optimal balance of sensitivity and specificity captures the most true positives and true negative patients, that is, the threshold where the number of true positives and true negatives is at its highest.

Contributors JOS devised the idea for the paper and also was the sole author.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient consent for publication Not required.

Provenance and peer review Not commissioned; internally peer reviewed.

© Author(s) (or their employer(s)) 2019. No commercial re-use. See rights and permissions. Published by BMJ.

To cite: O'Sullivan JW. *BMJ Evidence-Based Medicine* Epub ahead of print: [please include Day Month Year]. doi:10.1136/bmjebm-2019-111179

References

1. O'Sullivan JW, Albasri A, Koshiairis C, *et al*. Diagnostic test guidelines based on high-quality evidence had greater rates of adherence: a meta-epidemiological study. *J Clin Epidemiol* 2018;103:40–50.
2. López-López JA, Sterne JAC, Thom HHZ, *et al*. Oral anticoagulants for prevention of stroke in atrial fibrillation: systematic review, network meta-analysis, and cost effectiveness analysis. *BMJ* 2017;359:j5058.
3. Owen RK, Cooper NJ, Quinn TJ, *et al*. Network meta-analysis of diagnostic test accuracy studies identifies and ranks the optimal diagnostic tests and thresholds for health care policy and decision-making. *J Clin Epidemiol* 2018;99:64–74.
4. O'Sullivan JW, Heneghan C, Perera R, *et al*. Author Correction: Variation in diagnostic test requests and outcomes: a preliminary metric for OpenPathology.net. *Sci Rep* 2018;8:8354.